

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Depth gauge, Part Number 3823495, or equivalent
- Valve spring compressor tool, Part Number 3375182
- Cylinder head capscrew length gauge, Part Number 3163760
- Cylinder Head Leak Test Kit, Part Number 3164341.

Additional Service Items

- Straightedge
- Feeler gauge.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

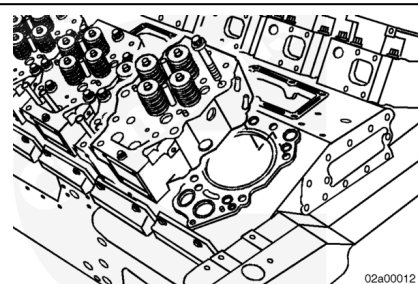
- Disconnect the batteries and air supply to the air starter to prevent accidental starting. See equipment manufacturer service information.

- Open the drain cock in the radiator bottom tank. Remove the lower radiator hose, if necessary.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/57/57-010-033.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/57/57-011-007.html)
- Remove the fuel manifold. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/57/57-006-013.html)
- Remove the aftercooler. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/57/57-010-002.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/57/57-003-011.html)
- Remove the rocker levers. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/57/57-003-009.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/57/57-006-026.html)
- Remove the rocker lever housing. Refer to Procedure 003-013 in Section 3.
(/qs3/pubsys2/xml/en/procedures/57/57-003-013.html)
- Remove the water vent lines from the head. Refer to Procedure 008-017 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-017.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



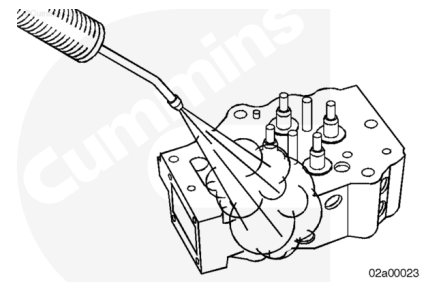
02a00012

Remove the seven capscrews and the cylinder head. Remove and discard the gasket.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



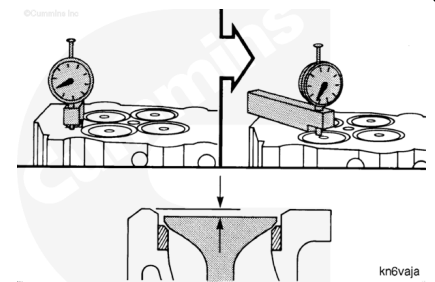
Clean the heads with solvent.

Check for broken springs or other damage.

Use a depth gauge, Part Number 3823495, or equivalent. Measure the valve head depth.

Note : If the valve depth exceeds specifications, the cylinder head must be rebuilt.

Note : If valve seats are being replaced, they must be machined after installation.



Allowable Valve Depth from the Combustion Face

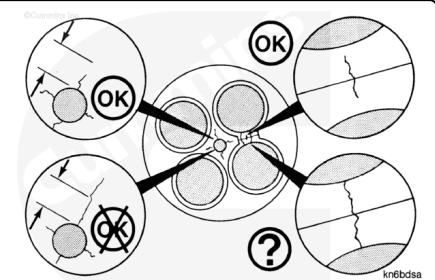
mm		in
0.00	MIN	0.000
0.70	MAX	0.028

Check for cracks on the combustion surface.

Note : If a crack around the injector bore exceeds the maximum length, the cylinder head must be replaced.

Allowable Crack Length

mm		in
6.35	MAX	0.25



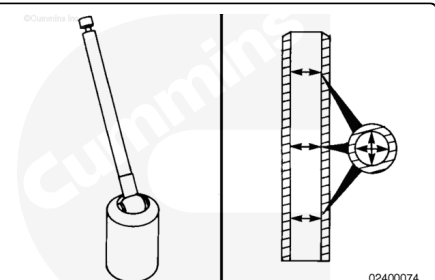
Note : Both ends of a crack between valves must be visible. If one end of a crack extends into the valve seat bore (behind the valve seat), the condition of the cylinder is questionable. To be sure that the cylinder head is reusable, remove the valve and valve seat.

Check the valve guide for chips and cracks.

Measure the inside diameter of the valve guide with a ball gauge or a dial bore indicator.

Valve Guide Inside Diameter (Installed)

mm		in
9.99	MIN	0.3933



Valve Guide Inside Diameter (Installed)		
mm		in
10.01	MAX	0.3941

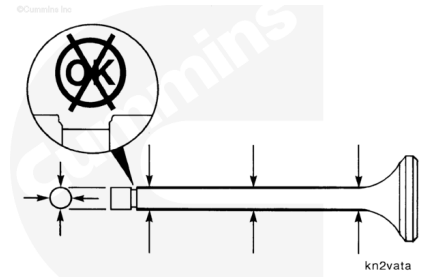
The first 13 mm [0.500 in] from the top or the bottom of the valve guide inside diameter can be too large and the valve guide can still be reusable. If the middle section, 13 mm [0.500 in] or greater from the top or bottom, is out of tolerance, the valve guide **must** be replaced.

It is **not** necessary to remove any black or brown stains on the valve stem, unless the buildup affects the movement of the valve in the valve guide. Use nothing more than a coarse crocus cloth (1000 grit) to remove the stains.

If there are scuffs or marks that can be felt with the fingernail, the valve **must** be replaced.

Check the collet groove area for wear. If the groove area is worn, the valve **must** be replaced.

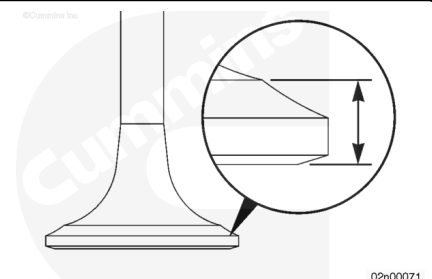
Measure the valve stem outside diameter.



Intake Valve Stem Outside Diameter		
mm		in
9.938	MIN	0.3913
9.957	MAX	0.3920

Exhaust Valve Stem Outside Diameter		
mm		in
9.908	MIN	0.3901
9.928	MAX	0.3909

Place the valve on a flat surface. Check the height of the outside diameter. If the height is less than 4.06 mm [0.160 in], the valve **must** be replaced.



Note : Cummins Inc. recommends using new valve springs when the cylinder head is reconditioned.

A bent or broken spring **must not** be used again.

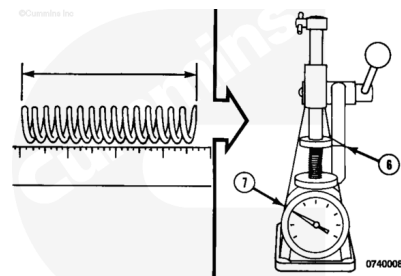
A spring with a notch worn in the first coil **must not** be used again. The ends of the springs will wear into the first coil, creating a worn notch in the material.

If a worn notch can be seen or felt, the valve spring **must not** be used again.

A spring that has these notches will make a clicking sound. Compress the spring until the end of the first coil is completely collapsed to check for a click sound. A large notch will cause a clicking sound.

Check both ends of the spring for the clicking sound.

Check the spring free length.



Measurements

	mm	in
Inner Valve Spring Approximate Free Length:	72	2.83

Measurements

	mm	in
Outer Valve Spring Approximate Free Length:	81.7	3.22

Use a valve spring compressor tool, Part Number 3375182, to check the spring force at the indicated test height.

Test Height			Valve Spring Force		
	mm	in	N		lbf
Inner	46.0	1.81	205	MIN	46.09
			227	MAX	51.03
	32.0	1.26	354	MIN	79.58
			392	MAX	88.13
Outer	51.0	2.01	402	MIN	90.37
			446	MAX	100.26
	37.0	1.46	666	MIN	149.72
			736	MAX	165.46

If the valve spring is **not** within specifications, the valve spring **must** be replaced.

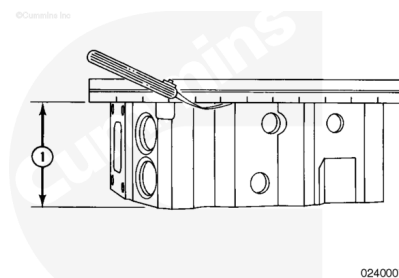
Use a straightedge to measure the flatness of the head gasket surface.

The head **must** be resurfaced if a feeler gauge larger than 0.09 mm [0.004 in] will fit between the straightedge and the cylinder head.

The head can be resurfaced as long as the head thickness measurement is within specifications.

Minimum Head Thickness (1)		
mm		in
104.65	MIN	4.120

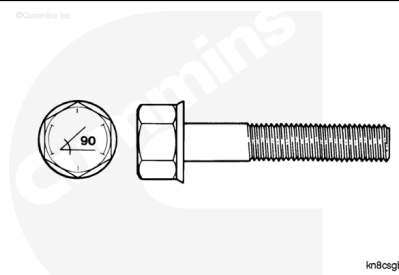
Note : If the head is resurfaced, be sure that the valve depth in the head is adjusted accordingly.



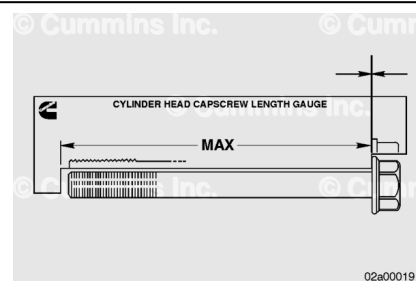
Tightening flange head capscrews with the torque-to-yield method of installation places the capscrew beyond the yield point and permanently stretches the capscrew. These capscrews can be reused throughout the life of the engine, unless the capscrew exceeds the specified free length.

The free length **must** be checked to avoid bottoming in the block during installation and to make sure adequate clamp load can still be maintained.

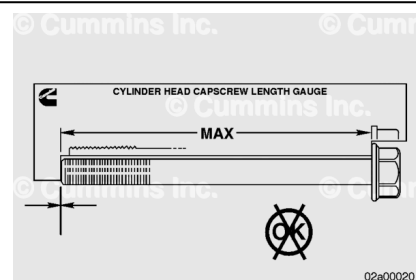
Use cylinder head capscrew length gauge, Part Number 3163760, to check capscrew free length.



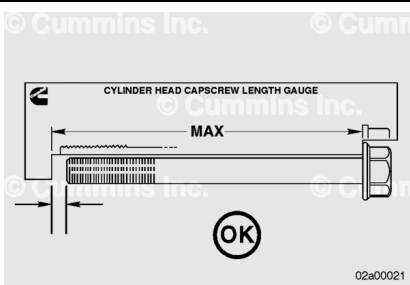
To check the capscrew free length, place the head of the capscrew in the slot with the flange against the base of the slot.



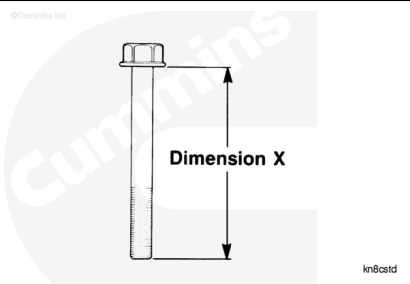
If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.



If there is clearance between the end of the capscrew and the bottom base of the gauge, the capscrew can be reused.



The capscrew can also be checked using a set of calipers. The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (dimension X).

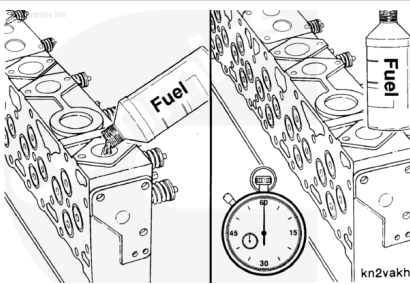


Cylinder Head Capscrew Free Length		
mm		in
163.80	MAX	6.449

Leak Test

Inspect the valves for indications of leakage or burning. If indications of leakage or burning are found, the valves and the seats **must** be resurfaced.

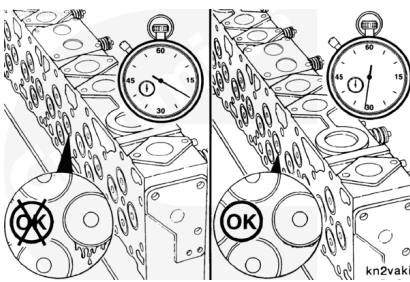
Test the cylinder head for damage. Set the head down with the exhaust ports facing up. Pour fuel into one of the exhaust ports until it is full. Start the timer.



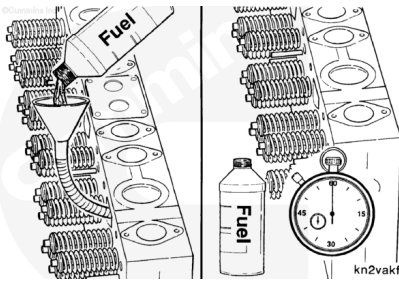
If a drop of fuel drips or runs down the face of the head within 30 seconds, the exhaust valves and seats **must** be resurfaced.

If a drop of fuel has **not** run down the face of the head in 30 seconds, the exhaust valves are reusable.

Repeat the process for all 12 cylinders.



Pour fuel into one of the intake ports until it is full. Start the timer.



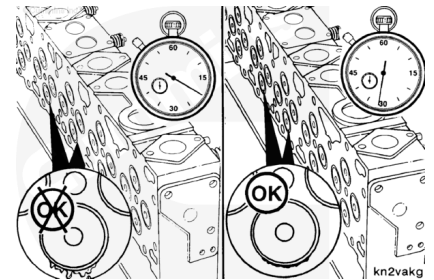
If a drop of fuel drips or runs down the face of the head within 30 seconds, the intake valves and seats **must** be resurfaced, and the intake valves replaced.

If a drop of fuel has **not** run down the face of the head in 30 seconds, the intake valves are reusable.

Repeat the process for all 12 cylinders.

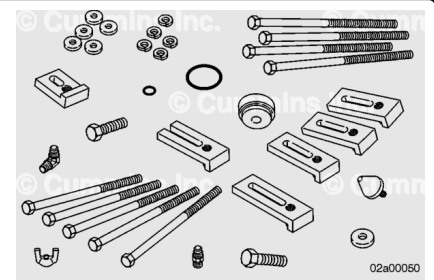
Note : If any of the intake or exhaust valves fail the test, all of the valves and valves seats **must** be resurfaced or replaced.

Note : Intake valves can not be reground.

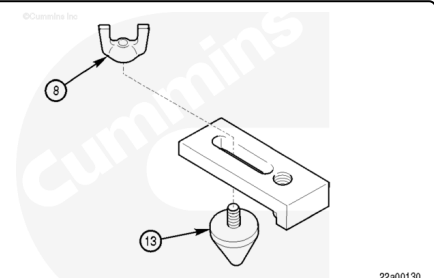


Pressure Test

Use the Cylinder Head Leak Test Kit, Part Number 3164341, to perform the Cylinder Head Hydrostatic Leak Test.



Assemble the wing nut (8), clamp, and plug (13), as necessary.



Note : The QST30 cylinder head, as shown, uses five clamp and plug assemblies; other models use **only** the

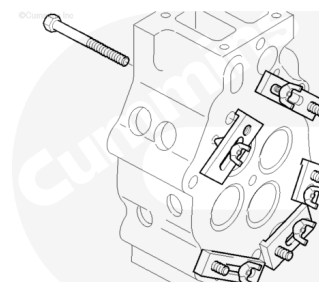
four longer assemblies.

Install the clamp and plug assembly into the cylinder head, blocking the coolant passages.

Insert the appropriate capscrew through the valve guide side of the cylinder head, and thread it into the clamp.

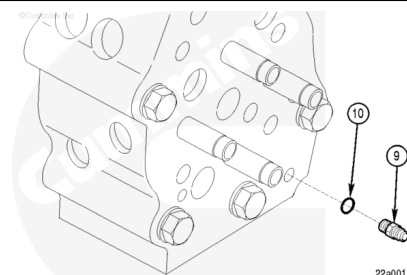
Tighten the capscrew.

Torque Value: 20 n•m [177 in-lb]



22a00131

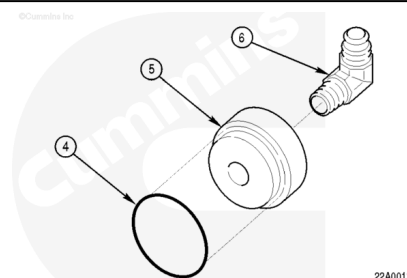
Install the sealing washer (10) and pipe nipple (9) into the coolant vent port of the cylinder head.



22a00132

Install the o-ring (4) into the o-ring groove of the test adapter (5).

Thread the adapter elbow (6) into adapter (5).

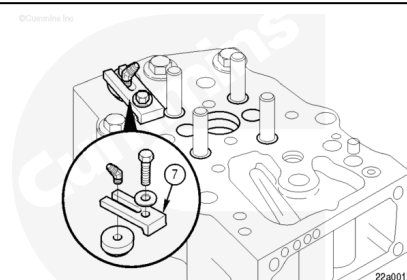


22A00133

Place the test adapter assembly into the cylinder head.

Install the clamp (7) over the test adapter assembly and secure it with the appropriate capscrew and washer.

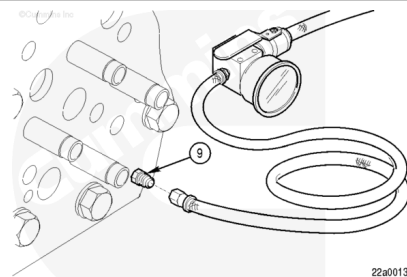
Torque Value: 54 n•m [40 ft-lb]



22a00134

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



22a00135

Connect the Pressure Regulator Valve Kit to the pipe nipple (9).

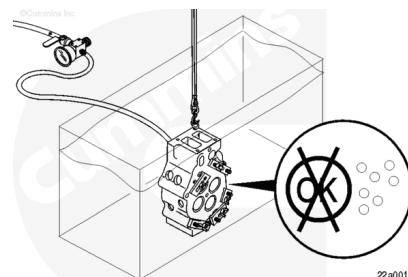
Connect the Pressure Regulator Valve Kit to shop air and adjust the air pressure.

Measurements

	kpa	psi
Air Pressure:	345	50

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Attach an eyebolt to the cylinder head.

Use a suitable lifting device to lift the cylinder head.

Immerse the cylinder head into a tank of water.

Check for bubbles. Bubbles indicate an air leak.

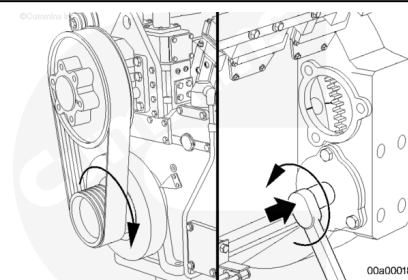
If the cylinder head leaks, it **must** be repaired or replaced.

Pressure Test

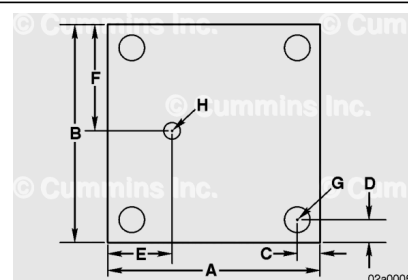
Intake Port

Note : This test can be carried out with the cylinder head installed or removed from the engine.

If the cylinder head is installed on the engine, make sure the intake valves are in the "closed" position on the cylinder head being checked. This can be done by barring the engine over. When the rocker lever is loose, this indicates the intake valves are closed.



Fabricate a plate to seal the cylinder head intake port as per the graphic and dimensions illustrated. A suitable pressure fitting **must** be installed at position "H" to pressurize the intake port with compressed air.

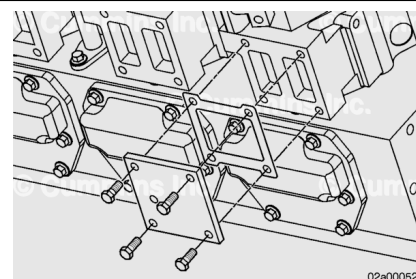


Reference	Dimension (mm)	Dimension [inch]
A	100	3.94
B	103	4.06
C	11	0.43

D	11	0.43
E	30	1.18
F	50	1.97
G	11	0.43
H	To fit Pressure fitting	To fit Pressure fitting
Plate thickness 5 mm [0.20 inch]		

Use four M10x25mm capscrews to install the plate and gasket to the intake port of the cylinder head.

Torque Value: 45 n•m [33 ft-lb]



⚠ WARNING ⚠

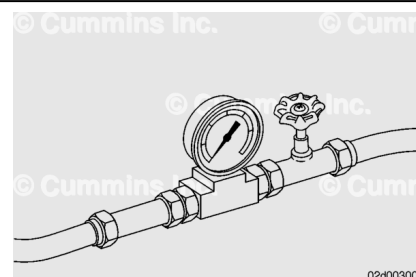
Do not exceed an intake pressure of 2 Bar [29.4 psi] or damage to the cylinder head can occur.

Note : If the intake port can not be pressurized due to excessive leakage, correct valve seating must be verified. If the valves are fully seated, perform the Cylinder Head Leak Test in this procedure.

Use a regulated air pressure line to pressurize the intake port to 2 Bar [29.4 psi].

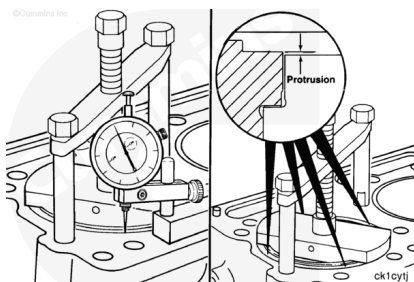
Check for leakage around the intake port or the push tube sleeve, use an appropriate leak detection fluid.

If leakage from the push tube sleeve is identified, replace the cylinder head.



Install

Measure the liner protrusion. Refer to Procedure 001-064 in Section 1 (/qs3/pubsys2/xml/en/procedures/57/57-001-064.html).



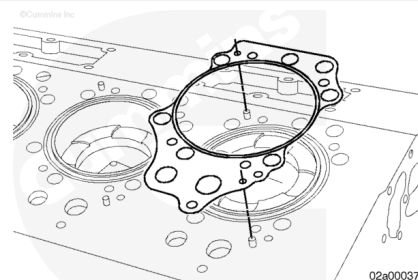
Inspect the cylinder and cylinder head. Make sure the mounting surfaces are clean and free of dust.

Oil the cam follower cups with clean engine oil.

Install the cylinder head gasket.

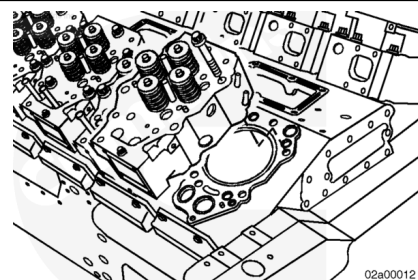
Note : Do not apply gasket sealant to the grommets. Grommets can swell from exposure to sealant.

Note : Make sure that the grommets are properly installed in the head gasket, and make sure the grommets seal properly over the push tube holes.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Lubricate the cylinder head mounting capscrew threads with new engine oil.

Apply anti-seize compound to the underside of the flange on the flange head capscrews.

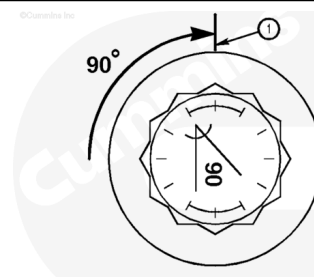
Install the cylinder head assembly and the cylinder head mounting capscrews.

Hand-tighten the cylinder head capscrews.

Tighten capscrews on one head (1 to 6). They **must** be rotated an additional 90 degrees in the direction of tightening.

Torque Value: 217 n·m [160 ft-lb]

After tightening the capscrews, mark the cylinder head and the 12-point capscrew head to provide a reference point for the additional turn.



Rotate the capscrew until the mark on the cylinder head is between the third and fourth points beyond the index mark.

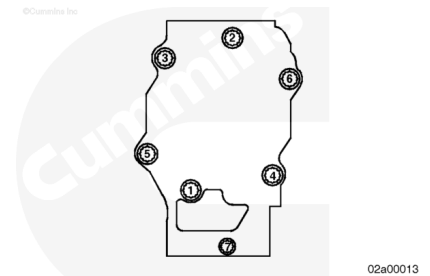
Tighten (M16) capscrews numbers 1 through 6 in sequence as shown.

Torque Value:

1. 149 n•m [110 ft-lb]
2. 217 n•m [160 ft-lb]
3. Tighten 90 degrees.

Tighten (M10) capscrew number 7.

Torque Value: 68 n•m [50 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To avoid arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the water vent lines to the head. Refer to Procedure 008-017 in Section 8. (</qs3/pubsys2/xml/en/procedures/57/57-008-017.html>)
- Install the banjo fittings and two brass washers to each hole.
- Tighten the banjo fittings.

Torque Value: 11 n•m [97 in-lb]

- Install the rocker lever housings. Refer to Procedure 003-013 in Section 3. (</qs3/pubsys2/xml/en/procedures/57/57-003-013.html>)
- Install the injectors. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/57/57-006-026.html>)
- Install the rocker levers. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/57/57-003-009.html>)
- Adjust the valves. Refer to Procedure 003-004 in Section 3. (</qs3/pubsys2/xml/en/procedures/57/57-003-004.html>)
- Install the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/57/57-003-011.html>)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11. (</qs3/pubsys2/xml/en/procedures/57/57-011-007.html>)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/57/57-010-033.html>)

- Install the fuel manifold. Refer to Procedure 006-013 in Section 6.
(/qs3/pubsys2/xml/en/procedures/57/57-006-013.html)
- Install the aftercooler. Refer to Procedure 010-002 in Section 10.
(/qs3/pubsys2/xml/en/procedures/57/57-010-002.html)
- Close the drain cock in the radiator bottom tank. Install the lower radiator hose, if necessary.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/57/57-008-018.html)
- Connect the batteries and air supply to the air starter. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/57/57-013-009.html)
- Operate the engine. Check for leaks.

Last Modified: 20-Sep-2022
